CTT infuses inefficiency into Gold futures trading: An empirical Analysis

By

Dr.Velmurugan.PS, Fulbright Fellow, Asst.Professor, Dept.of Commerce, Pondicherry University

&

Rinku Champramary, Doctoral Research Scholar, Dept.of Commerce, Pondicherry University

The rationale behind introduction of CTT by government of India is still unclear. But as per few experts, revenue generation was the topmost priority behind the imposition of CTT, besides checking price volatility. In this study by adopting the bootstrap, and modified GARCH methodology, we assessed the impact of CTT in non-agriculture futures commodities with special reference to gold futures traded in MCX. The study finds that CTT has increased the price volatility in gold futures and has significantly reduced the turnover in trading volume, weakening the market to great extent. The imposition of CTT not only affected the gold futures market but also contributed to the inefficiency of the market as the market has still not yet been able to digest the shock that it had received after the implementation of CTT. As per the revenue generation is concerned, if sole purpose for introducing CTT in the commodity futures market was to increase the revenue, than it is not suitable for development of commodity futures market, because CTT has already increased the transaction cost, and market players have already responded to it as volume came down approximately to 43% to 53%. We conclude that CTT has direct impact on policy aspirations as there is a need to transform Indian markets into 'price setter' from being 'price takers.

Key words: Gold futures, Commodity Transaction Tax, Price Taker, Fiscal Deficit, Bootstrap, MGARCH

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1. Introduction:

The levying of Commodity Transaction Tax (CTT) for non-agriculture commodities in India has been under debate since its inception. It is a derived version of securities transaction tax, which dates way back to Keynes 1936[1] and Tobin 1972[2]. Since then its importance has been growing and spreading gradually under various forms to different continents across the globe[3]. As transaction tax is of great importance to various stake holders, be it policy makers or the investors, the administration may like to raise more revenue for the government or to reduce the speculation, by imposing it. On the other hand such transaction tax may go against the investors and other market participants. Due to such conflicting interests among the stake holders, experts debate on its probable implications with supportive theories and empirical evidence to show the pros and cons of levying such financial transaction tax. Several of such proponents of financial transaction tax argue that transaction taxes will reduce the excess volatility through minimizing speculations and encouraging long term investment[4]. But, as well said, every merit has its own demerit. Many opponents also argue that such tax on financial transaction will bring down the securities values, liquidity etc. thus reducing market efficiency and driving the investors away to other offshore markets[5].

In India CTT was first proposed in the union budget 2008-09 @ of 0.017% but failed to succeed due to strong opposition from different point of view expressed by various stake holder. But such apprehension did not last long enough. Again in 28th February 2013, the then finance minister P Chidambaram came with fresh proposal in the union budget 2013-14, to levy @0.01 percent CTT for futures contract on non-agriculture commodities traded in Indian markets, applicable only to the sellers in the future contracts. Thus India finally succeeded in attaining a second place in the world next to Taiwan to impose commodity transaction tax without further validating the causes and consequences of commodity futures trading.

[1] John Maynard Keynes proposed STT in US to deal with volatility and speculators in foreign exchange and the stock market.

[2] James Tobin proposes currency transaction tax on spot conversion of currency to other currency in order to indirectly reduce the excess volatility.

[3] Proposed in Europe in 2011 to have a common system of FTT. Currency transaction taxes (CTT-Tobin Tax) levied by Brazil on foreign exchange.

[4] Proponents like Keynes, Tobin, Summers and summers, Stiglitz,(1989)

[5](Umlauf, 1993), tells that large part of trading activities migrated in response to introduction of such taxes from Sweden to London. (Campbell & Froot, 1994), study suggest that it may lead to reduction in trading via migration in foreign markets or by substituting their investment to other securities.

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The main proponents of commodity transaction tax in India Shri. P. Chidambaram's rationale behind introducing CTT was to bring commodity market on par with the securities market where a securities transaction tax is levied[6]. The rationale behind saying that there is "no distinction between securities and commodity derivatives trading' has already sideline the economic functions of commodity derivatives Viz. Price discovery and risk management and in doing so, is likely to deter commodity trading on Indian exchanges and shift transactions to lower cost exchanges overseas, or to illegal markets. As trade volume decreases, so too will any revenues generated by the CTT" (Charles M. Seeger, J.D, (2013).

Commodities further differ from securities because a "commodity exchange is clearly not a stock exchange" (Pavaskar & Ghosh, 2008) because of their difference in objectives. The main motive of the stock market players is to make profits from the rising stock values and investment in it. But this is not so in the case of commodity futures market because the main purpose in this market is to hedge the price risks in both the falling and the rising markets, depending on their commitments in the physical markets[7]. Thus opponents are of the view that the imposition of commodity transaction tax in futures markets in India will hamper the price discovery and hedging mechanism that are inherent characteristics of commodity futures market.

In this background this study tries to find out the effect of CTT on Indian non-agriculture commodity futures market, after the announcement of imposing (CTT) from 1st July 2013 onwards. The papers attempts to put forward the market behavior before and after the imposition of such (CTT), analyzing the changes in Trading Volume, Trading Lots, Market Return and market efficiency after the introduction of CTT in India.

The study is dealt in following way: Section 2 gives the overview of commodity futures market in India. Section 3 gives the review, Section 4 the data and methodology adopted, Section 5: explains the empirical findings. And Section 6 outline the conclusions and discussions.

2. Overview of Indian commodity futures market.

Though commodity trading dates way back to 19th century, its revival could be seen only in the new millennium with recognition granted to three national commodity exchanges, particularly MCX, NCDEX and NMCE. Of total 12 recognized commodity exchanges, six are national commodity exchange and six are commodity specific regional exchanges.

[6] In his Budget speech, Chidambaram stated, "There is no distinction between derivative trading in the securities market and derivative trading in the commodities market, only the underlying asset is different. It is time to introduce CTT in a limited way. Hence, I propose to levy CTT on non-agricultural commodities futures contracts at the same rate as on equity futures that is at 0.01 percent of the price of the trade. Trading in commodity derivatives will not be considered as a 'speculative transaction' and CTT shall be allowed as deduction if the income from such transaction forms part of business income."

[7] (Pavaskar & Ghosh, 2008)

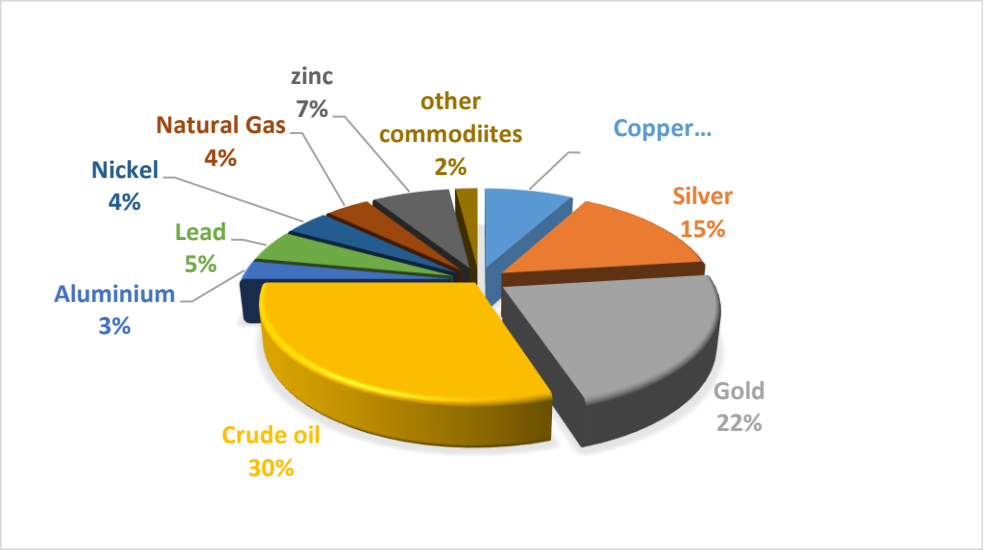
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Table 1: Total value and volume of trade in major exchanges.

Name of exchange	Volume of Trade (000 Tons)	Value (Rs crores)	% share (In value terms)
MCX	911,229	5,634,194	84.1%
NCDEX	217,737	1,019,588	15.2%
NMCE	6,028	29,368	0.4%
Total of National exchanges	**1,134,994**	**6,683,150**	99.80%
Regional exchanges	3,019	13,230	0.2%
Grand Total	**1,138,013**	**6,696,380**	**100.0%**

Note- Natural Gas Volumes are not included in the Total Volume
Source- SEBI Annual report 2015-16

Chart 1: Percentage share of Total value of commodities traded in MCX



Source: MCX Annual report 2015-16

The four national commodity Exchanges contributed 99.84 % of the total value of trade in the commodity futures market. These are MCX Mumbai (80.55 %), NCDEX Mumbai (18.75 %), NMCE Ahmedabad (0.53%) and ACE Mumbai (0.02%). review (April to June 2015), the total value of trade in all commodities traded at the recognized

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Exchanges was ₹16.86 lakh crore as against ₹ 16.09 lakh crore during the previous quarter (January to March 2015) and ₹14.55 lakh crore during the corresponding period.

(Bjursell, Wang, & Yau, 2012) in their paper says that a 0.02 percent transaction tax is big enough to swipe the whole market leaving no space for the government to raise no tax. The expectation of government of India is that CTT will bring revenue of 45 billion to the exchequer without looking at the side effects of it. It has already reflected as it has already created a panic like situation in the non-agriculture futures commodities, as the cost of transaction is increasing from Rs 183 to Rs 683 per crore. "As exchanges charge Rs 183 per crore as a transaction cost as against that CTT is Rs 1,000 per crore on the sales side which results into the increase in the impact cost and increases the cost of hedging transactions, as sellers have to pay 0.01 percent tax, that is, Rs 10 for a transaction value of Rs 1 lakh"[8]. So in such situation why is still the government in favor of imposing such tax which has no potential at all to generate more revenue.

3. Review of literature:

The aim of Imposing Financial Transaction taxes has its own purpose, some are to curve the particular issue like crisis etc. and some may be for general purpose which may or may not be achieved[9]. But, they are often viewed as a tool to reduce market volatility and is likely to be introduced in a period by those countries experiencing high volatility (Capelle-Blancard & Havrylchyk, 2014). Proponents of financial transaction tax like Keynes, Tobin, Stiglitz, Summers & Summers, Edgar Feige, give numerous reasons for proposing financial transaction tax; like curving volatility, speculations, more fair and equitable tax collection[10], feasible and less sensitive to tax avoidance and evasion[11] [12]. With such motives, various financial transaction taxes were proposed by experts. such as securities transaction tax proposed by Keynes for dealings on wall Street, currency transaction tax such as Tobin tax and Spahn tax proposed by James Tobin on all spot conversion of currency to another and Paul

[8] http://www.moneycontrol.com/news/commodities/budget-2015-16-expect-commodity-transaction-tax-to-be-removed-says-mcx_1313062.html?utm_source=ref_article

[9] En.Wikipedia.org

[10] The International Monetary Fund (IMF) delivered a report in 2010 titled "A Fair and Substantial Contribution by the Financial Sector" which made reference to a financial transaction tax as one of several options.

[11] Fraser Reilly-King of Halifax Initiative also points out that the key issue and advantage of an FTT is its relatively superior functional ability to prevent tax evasion in the financial sector.

[12] Economist Rodney Schmidt, principal researcher of The North-South Institute, also concurred that a financial transaction tax is more technically feasible than the "bank tax" proposed by the IMF in 2010.

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Bernd Spahn consisting of two tiers rate structure[13], automated payment transaction tax proposed by Edgar L. Feige in extension to the ideas of Keynes and Tobin which consist of flat rate tax on all transactions[14].

As we know every coins has two sides, so also the proponents has its opponents who have different views regarding the introduction of such taxes. Some of them are in the view that it will increase the transaction cost and bring down the liquidity in the market, thus reducing the values of securities traded (Amihud & Mendelson, 1991),(Kupiec & Sharpe, 1991), (Grundfest & Shoven, 1991), (Lo, Mamaysky, & Wang, 2000) also the turnover (Erickson & Wang, 2000),(Lindgren & Westlund, 1990),(Wang & Yau, 2012) in their study finds that such a tax will also reduce trading volume and increase price volatility and will only help in generating low tax revenues. (Baltagi, Li, & Li, 2006) find that an increase in the Chinese stamp tax rate from 0.3% to 0.5% reduces the trading volume by one third.

(Roll, 1984) examines whether countries with transaction taxes have a lower volatility, and finds the answer to be negative. Similarly, (Umlauf, 1993) finds that the introduction of, or increase in, the Swedish transaction tax would lead to an increase in stock market price volatility. Examining the effect of changes in transaction taxes in four Asian countries, (Hu, 1998)Hu finds insignificant impact of transaction taxes on price volatility, but significant negative impact on stock returns in some countries. (Saporta & Kan, 1997) investigate the effect of the UK stamp duty on the volatility of continuously compounded 15-minute returns over daily, weekly and monthly time periods. The results reveal that there is no evidence of any kind of effect, positive or negative, on the volatility.

There is also empirical evidence that imposition of /increase in STT would decrease return. Hu (1998) finds that upon the announcement of the tax, the average return fell 1.57 per cent in Taiwan and by 0.55 per cent in Korea. (Umlauf, 1993) estimates that the Swedish All-Share equity index fell 2.2 per cent when the 1 per cent tax was announced in 1983 and by 0.8 per cent when the tax rate was increased in 1986. In the UK, (Saporta & Kan, 1997) report that the equity index declined by 3.33 per cent on the announcement of an increase in stamp duty rate from 1 per cent to 2 per cent. When the stamp duty was decreased from 2 per cent to 1 per cent and then from 1 per cent to 0.5 per cent, the index increased by 0.558 per cent and 1.054 per cent respectively.

[13] Paul Bernd Spahn opposed the original form of a Tobin Tax in a Working Paper "International Financial Flows and Transactions Taxes: Survey and Options, concluding "...the original Tobin tax is not viable. This new form of tax, the Spahn tax, was later approved by the Belgian Federal Parliament in 2004. It has a two-tier rate structure consisting of a low rate financial transactions tax and an exchange surcharge at prohibitive rates.

[14] The foundations of the APT tax proposal—a small, uniform tax on all economic transactions—involve simplification, base broadening, reductions in marginal tax rates, the elimination of tax and information returns and the automatic collection of tax revenues at the payment source.

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Though Potential revenue and ease of collection from imposition of a transaction tax are attractive to governments, however, (Umlauf, 1993) reports that the decreased levels of trading in response to transaction tax resulted in decreased capital gains revenue in Sweden. Recently, a study by (Schulmeister, 2008), (Jurečková & Picek, 2007) estimates the transaction tax revenue of different countries. It is seen that the revenue from transaction tax varies between 0.01 for France to 1.7 for Spain, depending on the tax rate.

(Chou & Lee, 2002) provide interesting empirical evidence on the effect of the transaction tax on liquidity and market efficiency. They demonstrate that after the tax reduction on the Taiwan Futures Exchange (TAIFEX), the TAIFEX assumed a leading role over the Singapore Stock Exchange (SGX) in the price discovery process for the index futures contracts. Supporting Chou and Lee, (Hsieh, 2009)also notes that the information advantage of SGX has diminished as the TAIFEX lowered its transaction tax. Finally, in a review article(Habermeier & Kirilenko, 2003) by), they conclude that transactions taxes have a significant impact on the transformation of investor demands into transactions. Transaction taxes are found to delay the price discovery process, increase volatility, and reduce market liquidity.

Using futures market data, (Aliber, Chowdhry, & Yan, 2003) find a positive relationship between transaction costs and volatility and a negative relation between trading volume and transaction cost in the foreign exchange futures market for the British Pound, Japanese Yen and Swiss Franc (against the U.S. Dollar). (Constantinides, 1986),(Hegde & Miller, 1989) show that increases in transaction costs cause investors to reduce the frequency and volume of trading. Another study by (Frino, Stevenson, & Duffy, 1998) by 15% due to increase in transaction fee by $1 per contract for the case of Austria. Similarly, (Chou & Wang, 2006)found that following the reduction in transaction tax, the trading volume had increased and the bid-ask spread had decreased.

From the review of literature, it could be understood that both proponents and opponents of financial transaction tax had placed there strong arguments with empirical evidence. It is clear from the review that most of these researcher attempted to find its implication in securities transaction tax concerned with stock markets. There is few evidence from existing literature in the probable implication of transaction taxes of commodity market, which have a different purpose for its existence. Hence, to bridge the gap paper attempts to check the outcome of the imposition of CTT in non-agriculture futures market in India

4. Data and methodology:

The study is focused only on single commodity futures that is Gold Futures traded in MCX. Gold is the second largest commodity futures traded in the MCX which is next to crude oil and regarded as one of the most active non-agriculture commodity future contract that is driving the commodity futures market in India, but

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unfortunately is included in the purview of CTT. The data in this study covers from 1st January 2011 to 30th July 2016, covering a total of 1616 observations. First, the log return is calculated for the study $R_t = P_t/P_{t-1}$ for which P_t is the price at time t. Apart from this the paper also considers trading value, open interest, trading volume etc. The summary statics for the data is given below in Table: 1.

Chart 1: Bullion futures trading

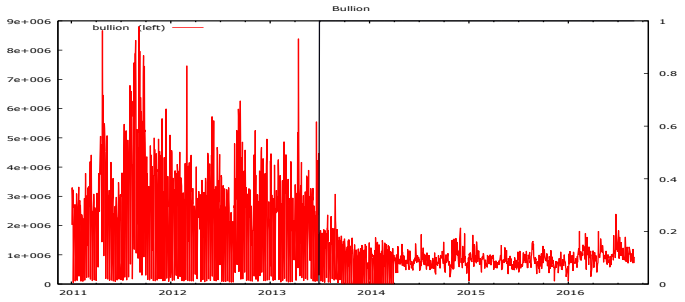


Chart 2: Agro futures trading

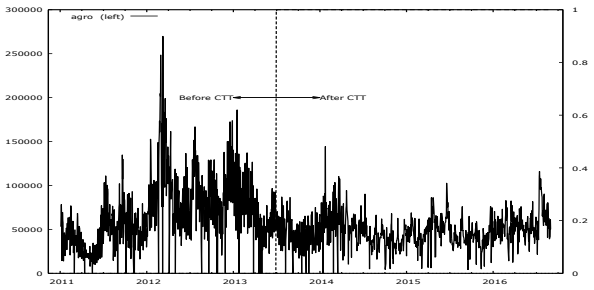


Table: 1 Before CTT

Year	Average trading value (in lots)	Average trading value in (lakhs)	Average Open Interest (In Lots)	Turnover Ratio (%)
2011	35100.39	875795.4	11289.2	310.92
2012	28324.21	839794.4	12308.85	230.1126
2013	33197.45*	937209*	13460.68*	246.6254

*indicate that average trading volume in lots from the month of 1st January to 31st July because this figure is before implementation of CTT

Table 2: After implementation of CTT

*indicate that average trading volume in lots from the month of July because this figure is after implementation

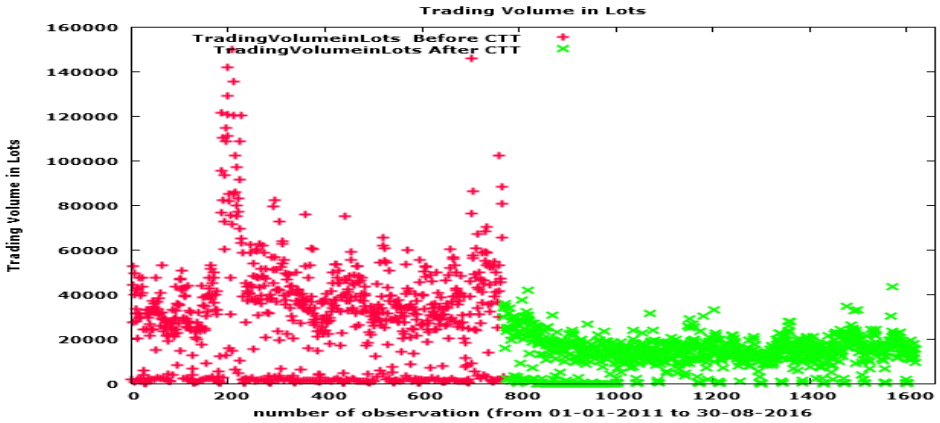
Year/month	Average trading value (in lots)	Average trading value in (lakhs)	Average Open Interest (In Lots)	Turnover Ratio in (%)
2013	16357.28*	479769.8*	8267.91*	197.8406
2014	12682.41	354085.8	8072.31	157.1101
2015	13324.57	351231.5	6623.306	201.177
2016	15794.23**	466291.4	7172.494	220.2055

of CTT.

**indicated up to the month of august 2016

Chart: 3 Trading Volume in Lots

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If we look at the above given Table:1 and Table: 2 we can clearly observe that there is a continuous decline in average trading value (in lots), average trading value in (lakhs), average Open Interest (In Lots) and Turnover Ratio. So what we can point out from this table is that the market is weakening. every year, signaling capital is moving out of the Gold futures market, which indirectly mean's the investors are hesitant to participate in this market partially attributed to the earlier discussed rise in transaction cost, that is additional burden imposed by the proposed mandatory cost in the form of CTT. This reflects that CTT is going against the long term policy implication of India that is to have one of the most developed exchange in the world which will act as a market leader, that is price setter rather than price taker, and to deliver the benefits of investments in gold from its physical form towards the paper format. But if we see the trading volume, open interest etc. CTT is dragging India backward way by increasing the transactions cost. That means the cost of investment in physical form is becoming more attractive than the paper form in (futures market) because of the increasing cost related issues while dealing in gold futures market.

Gold is an inevitable form of investment especially in a country like India which is treated auspicious, precious and currency. So the demand for gold is sure to rise in future for which the policy should be designed in such a way that demand does not lead to fiscal burden. In order to deal with such issues policy must be suitable for creating favorable avenues for investment in gold in the paper form rather than physical form like gold futures market to reduce the fiscal burden. But lack of suitable policy changes like imposition of CTT will only encourage the investors to invest in physical form, adding to India's fiscal burden.

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Table: 3 Summary Statistics

Statistics	Before CTT	After CTT	Total Observation
Number of Observations	766	850	1616
Mean	0.000323	0.000255	0.000286
Maximum	0.032066	0.057692	0.057692
Minimum	-0.082041	-0.053283	-0.082041
Std. Dev.	0.009421	0.010372	0.009933
Skewness	-1.558696	0.555558	-0.298181
Kurtosis	15.00305	8.077913	10.78755
MAD*	0.006057	0.007124	0.00636
Jarque-Bera	4908.510**	956.9505**	4104.900**

*denotes mean absolute deviation

** Significance levels denoting significant in this case.

The above table outlines the summary statistics of the daily return series of Gold futures traded in MCX.

The paper tries to investigate the economic impact of imposition of CTT from three different aspects of market quality parameters i.e. change in turnover (liquidity), market volatility, and market efficiency.

4.1: Trading Volume:

To examine the effect of commodity transaction tax and its impact on trading volume, we calculated the mean difference before and after the introduction of CTT. In order to deal with normality related issues, null hypothesis of equal trading volume the bootstrap method is employed (Baltagi et al., 2006) . Here, in this case, trading volumes were named as tv0 = (tv_1, tv_2..... tv_k) for k trading days before the imposition of CTT, and tv1 = ($tv1_1$, $tv1_2$..... $tv1_k$), for k trading days after the imposition of CTT. The following is the test statistic without assuming of equal variance between tv0 and tv1.

$$T(x) = \frac{\overline{tvo} - \overline{tv1}}{\sqrt{\frac{\sigma^2_{tv0}}{k} + \frac{\sigma^2_{tvi}}{k}}}$$

In order to attain the test statistics bootstrap distribution is adopted (Efron & Tibshirani, 1993). For which we need to do the following transformation $\widetilde{tv0}_i = tv0_i - \overline{tv0} + \bar{x}$ *and* $\widetilde{tv1}_i = tv1_i - \overline{tv1} + \bar{x}$ *where* $\overline{tv0}$ *and* $\overline{tv1}$ are the group means and $\bar{x}$ is the mean of both the sample. Then from n bootstrap sample($tv0^*$, $tv1^*$), in which (z*, y*), are the samples which are replaced after the result obtained from transformation that is $\widetilde{tv0}$ = ($\widetilde{tv0}_1, \widetilde{tv0}_2, \ldots \widetilde{tv0}_{k}^{z}$) and $\widetilde{tv1}$= ($\widetilde{tv1}_1, \widetilde{tv1}_2 \ldots \widetilde{tv1}_k$) respectively.

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$$T^*(x^*) = \frac{\overline{tv0}^* - \overline{tv1}^*}{\sqrt{\frac{\sigma_{tv0}^{2*}}{k} + \sigma_{tv1}^{2*}/k}}$$

Without making any assumptions on normality (we know that we don't have anything to worry about assuming equal variance[15] because we are considering here bootstrap hypothesis test) for comparing means across the groups for samples of size 2k, using block resampling, corresponding to z trading days before and z trading days after CTT, where Z=100, 150, 300, 550, and 767, in order to obtained the critical value of test statistic (T^*) is used.

4.2: Levene statistic for variance equality

In order to assess the changes in volatility we are comparing the variances of the returns before and after the introduction of CTT. We used levenes test for equality of variances (without assuming equal means across data groups) that is across m groups of data (in our case m=2)(Levene, 1960). The null hypothesis is $\sigma_1^2 = \sigma_2^2$ (variance assume to be equal) Let x_{ij} be the j[th] observation $(j = 1,2..n_i)$

In the ith group $(i = 1, 2)$, and do the transformation $z_{ij} = |x_{ij} - \bar{x}_i|$here $x_i = \sum_j x_{ij}/n_j$. The Levene statistic is the ratio of between-group to within-group variances forz_{ij}:

$$W = \frac{\sum_{i=1}^{2} n_i \, (\bar{z}_i - \bar{z}..)^2}{\sum_{i=1}^{2} \sum_j (z_{ij} - z_i)^2 / \sum_i (n_i - 1)}$$

$$\bar{z}_i = \sum_j z_{ij}/\, n_i \; and \; \bar{z}.. = \sum_i \sum_j z_{ij}/\sum_i ni$$

Under the assumption that $x_{ij}$is independently and normally distributed, W has an asymptotic $F(2, \sum_i (n_i - 1)$ distribution. However, the Levene statistic W is sensitive to non-normality, as pointed out by (Brown & Forsythe, 1974). The summary statistics of our return data are presented in Table 1. The Jarque–Bera test show strong evidence of non-normality of the returns. (Brown & Forsythe, 1974) suggest two modified Levene statistics obtained by replacing the arithmetic mean $\bar{x}_i$ with either a 10% trimmed mean or the median in the group. The modified Levene statistics are found to be robust against non-normality. We will present all the Levene statistics which includes originally proposed, non-parametric levene test, as well as median adjusted. To deal with the possible complication of non-normality and autocorrelation, we use the bootstrap with block resampling to assess

[15] (Efron & Tibshirani, 1993) in his book introduction to bootstrap "in considering a bootstrap hypothesis test for comparing two means, there is no compelling reasons to assume equal variance hence we don't make assumption's"

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the significance level of the test statistics obtained from a designed sample of 2k daily returns in the following manner. Corresponding to k days before and after the event, a sample of 2k daily returns is taken and the Levene statistics are calculated. To avoid arbitrariness of the sample length, we let k=100, 150, 300, 550, 767. Then, from within the above designed sample of 2k returns, a sample of 2k items is randomly drawn with block resampling and subsequently divided into two groups, with the first z items allotted to one group and the rest to another. Levene statistics are calculated for the bootstrap sample. This procedure is repeated 1,000 times to yield a bootstrap distribution of the Levene statistics. The significance levels of the Levene statistics are identified as the percentiles of the bootstrap distribution to which they (the statistics) correspond. For the clarification of bootstrap techniques we recommend reading (Efron & Tibshirani, 1993).

4.3: Modified GARCH model:

In this study, an autoregressive conditional variance or GARCH type model is going to be used, which is also used in previous studies like(Baltagi et al., 2006). (Engle, 1982) introduces the autoregressive conditionally heteroskedastic (ARCH) model, in which the conditional variance is a function of past squared errors. (Bollerslev, 1986) extends the model (GARCH) in the way that the conditional variance is a function of both past errors and past conditional variances. The GARCH (1, 1) model proposed by (Bollerslev, 1986) is

$$r_t = \mu_0 + \varepsilon_{tr}$$

$$var\ (\varepsilon_t | I_{t-1}) \equiv\ \sigma_t^2 =\ \alpha_0 +\ \alpha_1 \varepsilon_{t-1}^2 +\ \alpha_2 \sigma_{t-1}^2$$

Where I_{t-1} denotes the information set up to period t−1. Many extensions of the GARCH model have been extensively used to capture different aspects in financial time series. For further reference (Bollerslev, Chou, & Kroner, 1992)and (Bera & Higgins, 1993). The below given equation of modified GARCH model allows for regime switching in both the mean equation and the conditional variance equation:

$$r_t =\ \mu_0 + \mu_1 D_t +\ \varepsilon_t$$

$$var(\varepsilon_t | I_{t-1}) \equiv\ \sigma_t^2 =\ \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \alpha_2 \sigma_{t-1}^2 + \alpha_3 D_t + \alpha_4 D_t \varepsilon_{t-1}^2 + a_5 D_t\ \sigma_{t-1}^2$$

$$D_t = \begin{bmatrix} 0\ if\ 1 \le t \le t^* - 1 \\ 1\ if\ t^*\ \le t \end{bmatrix}$$

Where t^* is the switching point, in this study it is the date of introduction of commodity transaction test i.e July, 1, 2013 which is used as switching point. The modified GARCH allows volatility shocks to persist over time. Using dummy variables it also captures structural shift in the level and slope of time varying volatility. If no structural change occurs, the dummy coefficients will be insignificant. The dummy in the mean equation suggests a change in the mean level of returns as a result of the CTT while the dummies in the conditional variance equation

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suggest regime switching in the autoregressive structure. Under the assumption of error normality, the log likelihood function of the above modified model is: $\mathcal{L} = \sum -\frac{1}{2}\left(\boldsymbol{ln}(\boldsymbol{2\pi}) + \boldsymbol{ln\sigma_t^2} + \frac{\varepsilon_t^2}{\sigma_t^2}\right)$

We will use the BHHH (1974) algorithm to maximize the above log-likelihood function. As suggested by (Lee & Ohk, 1992) and (Li, Lin, & Li, 1997), the structural change in conditional variance equation might suggest a change in market efficiency. In the GARCH model, σ_{t-1}^2, conditional variance at time t−1, is a function of all past squared errors,$\varepsilon_{t-2,}^2$ ε_{t-3}^2 ... Negative α_4 and positive α_5 suggest that, after the event, the effect of ε_{t-1}^2 on σ_t^2 relatively decreases, but the effect of σ_{t-1}^2 which is a function of $\varepsilon_{t-2}^2\varepsilon_{t-3\ldots}^2$ on σ_t^2 relatively increases. This implies that volatility shocks are less quickly reflected and digested in the financial markets after the event, i.e., the markets become less efficient. Similarly, positive α_4 and negative α_5 suggests increasing effect of $\varepsilon_{t-1}^2$and decreasing effect of $\varepsilon_{t-2}^2\varepsilon_{t-3\ldots}^2$ on σ_t^2 which implies a more efficient market. If α_4 and α_5 have the same sign, one can probably say little about the market efficiency.

The similar modified GARCH model which allows regime switching in the variance equation, is also used by (Lee & Ohk, 1992) to study the effect of stock index futures listing on the underlying index. They find that the listing of the index futures has a significant effect on the cash market. Also, that stock market volatility increases significantly after the introduction of the index futures. (Li et al., 1997) also uses the same model to study the impact of settlement time on the market behavior. Their result shows that when the market employ a longer settlement time, the markets become less volatile and less efficient as well.

5.Results of the Investigation:

5.1. Trading Volume: The aim is to assess the changes in gold futures turnover, that whether there is any changes in the mean of turnover after the imposition of commodity transaction tax in non-agriculture commodities particularly Gold. Before and first of July 2013. In order to assess the changes by comparing the mean difference before and after the implementation of CTT for different trading period ranging from ± 100 to ±766 (before and after). The methodology adopted for generating the test statistic is shown in the methodology part. We adopted one of the most robust method to deal with our data as the data we are using is likely to suffer from non-normality and lack of equal variance across the group. So, under such circumstance the method that we adopted is best fitted. The results presented in Table: 4 shows the mean level of turnover before the imposition of CTT and after the imposition of CTT. From the table we can see that there is a change in the mean of before and after the implementation of CTT. The mean of trading volume after the implementation of CTT is lower than the mean of trading volume before CTT. The *t* statistics result for all the samples that are taken are significant at 95 percent

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confident level. Which gives us the strong message that there is a significant change in trading volume after the introduction of CTT in Gold futures trading value. The trading volume came down by approximately 43 to 53 percent. From this we can understand that the market players in commodity gold futures actively responded to the imposition of CTT by reducing their trade dealings from such commodities.

Table: 4. Trading Volume (in Rs. lakhs)

	Time Period*				
	± 100	± 150	± 300	± 550	± 767
Mean Before CTT	972451.262500	938223.856133	883466.781367	903498.503618	873603.152855
Mean After CTT	545088.067900	486217.437067	410732.331800	387819.276582	391161.200261
Ratio**	0.56053	0.518232	0.46491	0.429242	0.447756
t statistics	5.617(.001)	7.838 (001)	12.743 (001)	18.874(000)	19.065(000)

*Time period is nothing but samples taken for the study, in this case from 0 to 767 trading days.
**The ratio is the ratio calculated for mean volume after CTT over Before CTT.
Figure in the parentheses are significance levels calculated by bootstrapping method with block resampling.

5.2. Market Volatility:

Assessing the impact of CTT on return volatility is considered by using the variance test, for which the detailed methodology adopted are given in section above. Before going ahead, let us look and recall back the summary statistics given in the Table: 3, where the kurtosis is very high. This gives us an indication that that the daily returns series is having a fat tail distribution meaning that non-normality is the issue. From the result given under Table: 5 looking at the standard deviation for the subsamples before and after the introduction of CTT, it is observed that the standard deviation obtained after the introduction of CTT are larger than the ones assessed before CTT.

Table: 5 Levene Statistics

	Time Period				
ststistics	± 100	± 150	± 300	± 550	± 767
Standard Deviation Before CTT	.0134301	.0114896	.0090782	.0091554	.0094212
Standard Deviation After CTT	.0163190	.0147245	.0120026	.0106318	.0104991
Original Levene	4.227(0.41)	7.270(.007)	10.843(.001)	9.190(.002)	9.768(.002)
NP LT	4.072(.045)	6.729(.010)	6.621(.010)	9.260(.002)	15.907(.000)
Median-Adjusted	3.914(.049)	7.095(.008)	11.058(.001)	9.436(.002)	9.820(.002)

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Apart from this, the Table: 5 has result of three levene statistics like original levene test, Non-parametric levene test, Median adjusted is also depicted in the table along with the significant level obtained by using bootstrap method. As most of the statistics are found to be significant, it can be concluded that the market tends to be more volatile after the introduction of CTT, opposite to what was intended by its proponents in markets like Europe and U.S

5.3. Market Efficiency

The modified GARCH model defined in methodology part are estimated for the gold futures return. The results are presented in Table: 6., the modified GARCH model allows structural change both in the mean equation as well as the conditional variance equation. The bad news is that the imposition of CTT has a negative impact on the rate of return.

Table: 6 modified GARH result:

SL/NO.	μ_0	μ_1	α_0	α_1	α_2	α_3	α_4	α_5	$\mathcal{L}$
Normal GARCH RESULT	0.000139 (0.5023)		2.01E-06 (0.000)	0.071532 (0.000)	0.911222 (0.000)				5046.48 1.16092
Modified GARCH results.	0.000275 (0.2876)	0.000320 (0.46820)	9.13E-05 (0.3297)	0.07503 (0.000)	0.896272 (0.0000)	9.94E-06 (0.0107)	-0.000175 (0.0327)	.00326 (0.3176)	5176.3

For the modified GARCH model, before the event day, the persistence of volatility shocks, measured by α1 + α2, is 0.97 for gold return before CTT. After the event day, the persistence of volatility shocks, measured by $\alpha_1 + \alpha_2 + \alpha_4 + \alpha_5$, is 0.99. The dummy Coefficients in the conditional variance are all significant. α_4 is significantly negative and α_5 is significantly positive. These suggest that there is a regime switching in the conditional variance equation. As we discussed in the last section, a negative α_4 and a negative α_5 which means that we cannot speak much about the market efficiency. But the effect of σ_{t-1}^2 which is a function of $\varepsilon_{t-2}^2 \varepsilon_{t-3\ldots}^2 on\ \sigma_t^2$ relatively increases. This implies that volatility shocks are not efficiently assimilated in the Gold futures market after the introduction of CTT. Which means that it has significantly reduced the efficiency of risk management in the market in post CTT era. Thus negatively impacting the market efficiency. Introducing CTT has increased the

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transaction cost and discouraged the traders. Due to which, the market fails to respond to the arrived information as promptly as before, which leads to a less efficient market. This study shows that market has become more volatile as the information shocks are less quickly assimilated. In (Li, Lin, & Li, (1997), the market becomes less volatile and less efficient as the settlement time becomes longer. In French et al. (1987, Section 2.2), the GARCH (1,2) model is estimated separately for the whole sample period as well as for two sub periods for the daily S and P index. They find some evidence that the parameters are not stable across the sub periods in their sample. Their model does not allow inference on the stability of the parameters. However, the model in this study explicitly allows the parameters to change in sub periods by introducing dummy variables. From this new model we are able to make inference on the change and see whether the change is significant. The results reveal that, after the introduction of CTT, the immediate lagged error has less impact on the conditional variance, which implies that the Gold Futures market reacts less promptly to the information and therefore becomes less efficient.

Conclusions:

Several countries which adopted financial transaction tax such as STT have witnessed its negative effects on the growth and development of the financial markets. Results of this paper shows that India too had been experiencing such effects, emphasizing the need to act accordingly to ensure that the CTT is repealed and the commodity futures market is allowed to function in a fair and orderly manner. The results of the study reveals that, contrary to what the common proponents of transaction taxes claim such as (Summers & Summers, 1989), the market volatility witnesses an increased after CTT has been introduced in India's gold futures market. Imposition of CTT has affected all the stakeholders in non-agriculture commodity futures market, particularly in gold futures evident from the declining trend in open interest, trading value, turnover ratios etc. The comparative chart of Bullion and agro- commodity trading value depicts the heat the bullion commodity is facing because of CTT. The Gold Futures market had witness decrease in turnover in trading value indirectly affecting the liquidity, also increasing volatility in gold futures return which is consistent with findings in (Umlauf, 1993) and (Jones & Seguin, 1997) etc. This shows that CTT is not desirable or is contrary to the policy objectives of existence of healthy futures trading in commodities and its potential impact on policy aspirations in transforming Indian markets into 'price setter' from being 'price taker. If the government sole motive of introducing commodities transaction tax was to generate more revenue to its exchequer than we think that it is a big blunder. If we look at the history of financial transaction tax right from the congressional budget estimates in U.S in 1990s and estimates in japan we can see

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an over estimation effect by the proponents between the amount expected and actually collected as revenue without considering the importance of trading volume(Wang & Yau, 2012) to support the project tax collection estimates. So without any further delay this paper recommend the policy makers to reconsider the decisions regarding the impositions of CTT, in non-agriculture commodities futures.

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